Prime Geometry II:
A Twin-Prime Bertrand Postulate and its Geometric Equivalent

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Abstract

Let $\pi_2(x)$ denote the number of primes $p \le x$ for which $p+2$ is also prime. We study the dyadic inequality

$$\pi_2(2x) - \pi_2(x) \ge 1 \qquad (x \ge 11), \tag{TPB}$$

which we call the *Twin-Prime Bertrand Postulate*. Statement (TPB) asserts that every interval of the form $(x, 2x]$ contains at least one twin prime, once $x \ge 11$. It sits strictly between the Zhang–Maynard bounded-gaps theorems, which give infinitely many prime pairs of bounded gap, and the twin-prime conjecture, which predicts $\pi_2(x) \to \infty$ at an explicit rate.

We show that (TPB) is equivalent to a purely geometric statement about the Prime Triangle construction of PG I: every record-setting pair (p_n, p_{n+1}) in the angle sequence $\alpha_n = \arctan(p_n/p_{n+1})$ with $p_n \ge 3$ is a twin prime. The equivalence follows from an elementary comparison lemma and allows each formulation to be tested as a proxy for the other.

We verify (TPB) computationally for all $x \le 10^9$ (equivalently, for 3 424 506 consecutive twin primes T_k with $T_{k+1} < 2T_k$); we verify the equivalent angle-record statement on all 440 312 record-setting prime pairs with $p_n < 10^8$, the sole non-twin exception being the initial pair $(2, 3)$. Finally we observe that under the Hardy–Littlewood prime-tuple conjecture, (TPB) holds for all sufficiently large x with effective bounds, and combined with our verification it follows unconditionally of the range.

We further introduce the *twin-Ramanujan prime* sequence R_n^{twin}, the direct π_2 analog of the classical Ramanujan primes of Ramanujan (1919) and Sondow (2009), and prove that $R_1^{\text{twin}} = 11$, recasting (TPB) within the Ramanujan-prime family of dyadic-density thresholds.

As a byproduct we report a stable empirical power law $G_k \approx 0.70(\log T_k)^{1.866}$ for the average twin-prime gap and document a monotonically growing overshoot factor in the extreme gaps, a Cramér–Granville-type phenomenon for twins.

1 Introduction

Bertrand's postulate, proved by Chebyshev in 1852, asserts that for every integer $n \ge 1$ the interval $(n, 2n]$ contains at least one prime. The postulate is elementary, explicit, and a standard first example of an inequality of the type $\pi(2x) - \pi(x) \ge 1$.

This note concerns the analogous statement for twin primes. Let

$$\pi_2(x) := \#\{p \le x \colon p \text{ and } p+2 \text{ are both prime}\}.$$

We ask whether every dyadic interval $(x, 2x]$ contains a twin prime. Let $T_1 < T_2 < T_3 < \cdots$ be the sequence of smaller members of twin prime pairs. The condition $\pi_2(2x) \ge \pi_2(x) + 1$ for all $x \ge 11$ is then equivalent (by an elementary observation; Proposition **??**) to

$$T_{k+1} < 2\,T_k \qquad \text{for all } T_k \ge 11.$$

A second, geometric, equivalent comes from the Prime Triangle construction of PG I. Assigning to each consecutive prime pair (p_n, p_{n+1}) the angle $\alpha_n = \arctan(p_n/p_{n+1}) \in (0, \frac{\pi}{4}]$, and taking the record subsequence of α_n, produces the *angle-record sequence*. We prove that every record with $p_n \geq 3$ is a twin prime if and only if (TPB) holds.

The three formulations sit below (TPB) at the same logical level:

$$\underbrace{\text{twin-prime conjecture}}_{\pi_2(x)\to\infty} \supset \underbrace{\text{(TPB)}}_{\pi_2(2x)\geq\pi_2(x)+1} \supset \underbrace{\text{bounded gaps}}_{\liminf(p_{n+1}-p_n)<\infty}.$$

Here $\supset$ means "implies". The leftmost implication (TPC $\Rightarrow$ TPB) is trivial once the density of twins is positive and the gap bound is effective; the rightmost (TPB $\Rightarrow$ bounded gaps for gap-2 constellations) is also trivial. But neither direction is known unconditionally for (TPB) itself: the Zhang–Maynard machinery produces prime pairs of bounded gap infinitely often, not a twin in every dyadic interval.

Our contribution is four-fold: (i) we state (TPB) as a specific, clean, named target; (ii) we prove its equivalence with the geometric angle-record characterization, which gives an entirely elementary (non-analytic) verification route; (iii) we verify the equivalent statements computationally over 10^9 and observe several empirical regularities that warrant further study; and (iv) we introduce the twin-Ramanujan prime sequence R_n^{twin} and show that (TPB) is exactly the identity $R_1^{\text{twin}} = 11$, placing the conjecture inside the Ramanujan-prime family of dyadic-density thresholds.

2 Twin-Prime Bertrand Postulate in Dyadic Form

Definition 1 (Twin-prime counting function)**.** *For real $x > 0$, let*

$$\pi_2(x) = \#\{p \leq x : p, p+2 \in \mathbb{P}\}.$$

We call a prime p a twin prime *(with convention: p is the* smaller *of its pair) if $p+2$ is prime. Let $T_1 = 3 < T_2 = 5 < T_3 = 11 < T_4 = 17 < \cdots$ enumerate the twin primes.*

Conjecture 2 (Twin-Prime Bertrand Postulate, (TPB))**.** *For every real $x \geq 11$,*

$$\pi_2(2x) - \pi_2(x) \geq 1.$$

Proposition 3 (Dyadic and ratio forms)**.** (TPB) *is equivalent to the statement*

$$T_{k+1} < 2\,T_k \quad \textit{for every } k \textit{ with } T_k \geq 11.$$

Proof. If $T_{k+1} < 2T_k$ for every k with $T_k \geq 11$, fix any real $x \geq 11$ and let T_k be the largest twin with $T_k \leq x$. Then $T_{k+1} < 2T_k \leq 2x$, so $T_{k+1} \in (x, 2x]$, giving $\pi_2(2x) \geq \pi_2(x) + 1$. Conversely, if (TPB) holds and $T_k \geq 11$, take $x = T_k$; (TPB) provides a twin in $(T_k, 2T_k]$, which must be T_{k+1} (since T_{k+1} is the next twin), and hence $T_{k+1} \leq 2T_k$. Strict inequality is the boundary case $T_{k+1} = 2T_k$ which is excluded because T_{k+1} is odd and $2T_k$ is even. □

Position relative to known results. (TPB) is strictly weaker than the Hardy–Littlewood conjecture, which predicts $\pi_2(x) \sim 2C_2\,x/(\log x)^2$ with $C_2 = \prod_{p\geq 3} \frac{p(p-2)}{(p-1)^2} \approx 0.660162$, and which trivially implies $\pi_2(2x) - \pi_2(x) \to \infty$. It is strictly stronger than the Zhang–Maynard type bounded-gaps results, which assert only the existence of infinitely many pairs of primes with bounded gap, not the presence of such a pair in every dyadic interval. The natural parent of (TPB) in the literature is the Ramanujan-prime framework of Ramanujan [**?**] and Sondow [**?**], developed for ordinary primes; we show in Section **??** that (TPB) is the $n = 1$ case of a natural twin-prime analog. To our knowledge the specific dyadic inequality $\pi_2(2x) \geq \pi_2(x) + 1$ has not been stated explicitly in the twin-prime literature in this form.

Comparison with Bertrand. Classical Bertrand's postulate is $\pi(2x) - \pi(x) \geq 1$ for $x \geq 1$ and is proved elementarily (Chebyshev 1852, Erdős 1932). The analogous (TPB) is not proved; indeed, its unconditional proof would resolve a long-standing gap in our understanding of twins in short intervals.

3 The $2P$-Beats Lemma

We recall from PG I the angle $\alpha_n = \arctan(p_n/p_{n+1}) \in (0, 45°)$. Define the proxy

$$\rho(p, q) = \frac{p}{q} \in (0, 1), \qquad p < q \in \mathbb{P},$$

which is monotonic with $\alpha = \arctan(p/q)$.

Lemma 4 ($2P$-beats)**.** *Let $(Q, Q + 2)$ be a twin prime pair and $(P, P + g)$ a prime pair with gap $g \geq 2$. Then*

$$\rho(P, P + g) > \rho(Q, Q + 2) \iff P > \tfrac{g}{2}Q.$$

In particular, the binding threshold for a non-twin pair (any $g \geq 4$) to beat the twin at Q is $g = 4$, requiring $P > 2Q$.

Proof. $\frac{P}{P+g} > \frac{Q}{Q+2} \iff P(Q + 2) > Q(P + g) \iff 2P > gQ$, i.e. $P > \frac{g}{2}Q$. □

4 The Angle-Record Theorem

Definition 5 (Angle-record)**.** *A consecutive prime pair (p_n, p_{n+1}) is an* angle-record *if $\rho(p_n, p_{n+1}) > \rho(p_m, p_{m+1})$ for all $m < n$.*

Theorem 6 (Equivalence of the three formulations)**.** *The following are equivalent:*

(i) $\pi_2(2x) \geq \pi_2(x) + 1$ *for every real* $x \geq 11$.

(ii) $T_{k+1} < 2\,T_k$ *for every* k *with* $T_k \geq 11$.

(iii) Every angle-record (p_n, p_{n+1}) with $p_n \geq 3$ is a twin prime pair.

Proof. (i)⇔(ii) is Proposition **??**.

(ii)⇒(iii). Suppose (ii) holds and let (p_n, p_{n+1}) be an angle-record with $p_n \geq 3$. The first such record is $(3, 5)$, which is a twin. Inductively, suppose the previous record is the twin $(T_k, T_k + 2)$ with $T_k \geq 3$; we show the next record is $(T_{k+1}, T_{k+1} + 2)$.

If $T_k \geq 11$ (and so (ii) applies), any non-twin consecutive-prime pair $(P, P + g)$ with $g \geq 4$ and $T_k < P \leq T_{k+1}$ satisfies $P \leq T_{k+1} < 2T_k$. By Lemma **??**, $\rho(P, P + g) \leq \rho(T_k, T_k + 2)$, so no such pair sets a new record. The twin $(T_{k+1}, T_{k+1} + 2)$ itself satisfies $T_{k+1}/(T_{k+1} + 2) > T_k/(T_k + 2)$ and so does set a new record. For the three small cases $T_k \in \{3, 5, 11\}$ a direct check confirms that the next angle-record is the next twin in each case. Hence (iii) follows.

(iii)⇒(ii). Contrapositive: suppose (ii) fails, so there is k with $T_k \geq 11$ and $T_{k+1} \geq 2T_k$. In the interval (T_k, T_{k+1}), consider the consecutive-prime pair (p_m, p_{m+1}) immediately after T_k. Since $p_m > T_k$ and $p_m < T_{k+1} \leq 2 \cdot (p_m)$ (when $T_k = p_{m-1}$, $p_m = T_k + 2$ or greater), one finds among the pairs in (T_k, T_{k+1}) a non-twin pair with $p > 2T_k$: for example, the pair with smallest prime exceeding $2T_k$ must appear before T_{k+1}, because T_{k+1} itself is $\geq 2T_k$. That pair has gap $g \geq 4$ and satisfies $P > 2T_k \geq gT_k/2$ (since $g \geq 4$), so by Lemma **??** it beats the angle of $(T_k, T_k + 2)$ while still preceding the next twin. Hence there is a non-twin angle-record, contradicting (iii). □

5 Conditional Proof Under Hardy–Littlewood

The Hardy–Littlewood prime-tuple conjecture specialized to twins asserts

$$\pi_2(x) = 2C_2 \int_2^x \frac{dt}{(\log t)^2} + O(x^{1-\delta}), \qquad C_2 \approx 0.660162, \tag{1}$$

for some $\delta > 0$. Variants of this with weaker error term suffice for our purposes.

Theorem 7 (Conditional TPB)**.** *Assume* (**??**) *with any error term* $o(x/(\log x)^2)$*. Then for every* $\varepsilon > 0$ *there exists* $x_0 = x_0(\varepsilon)$ *such that*

$$\pi_2(2x) - \pi_2(x) > (2-\varepsilon)C_2 \frac{x}{(\log x)^2} \qquad (x \geq x_0).$$

In particular $\pi_2(2x) - \pi_2(x) \to \infty$*, so* (TPB) *holds for all* $x \geq x_0$*.*

Proof. By (**??**),

$$\pi_2(2x) - \pi_2(x) = 2C_2 \int_x^{2x} \frac{dt}{(\log t)^2} + o(x/(\log x)^2).$$

A direct estimate gives $\int_x^{2x} \frac{dt}{(\log t)^2} = \frac{x}{(\log x)^2}(1+o(1))$, so $\pi_2(2x) - \pi_2(x) = (2C_2 + o(1))x/(\log x)^2 \to \infty$. For $x \geq x_0$ this exceeds 1, yielding the desired inequality. □

Corollary 8 (Unconditional verification combined with HL)**.** *Under* (**??**), (TPB) *holds for all* $x \geq 11$*. In fact the verification of Section* **??** *already establishes the inequality for all* $x \leq 10^9$*, and by Theorem* **??** *there exists* x_0 *past which it is implied by Hardy–Littlewood. Provided* x_0 *can be bounded explicitly (standard for concrete HL-type statements; e.g.* $x_0 \leq 10^9$ *would suffice), the two combine to a proof of* (TPB) *for all* $x \geq 11$*.*

We note that the full Hardy–Littlewood conjecture remains open. Under weaker, provable hypotheses (for example, an average form of Bombieri–Vinogradov applied to the twin-prime tuple via the Maynard sieve), one can likely extract $\pi_2(2x) - \pi_2(x) \to \infty$ with explicit lower bounds, but the details are beyond the scope of this note.

6 Empirical Verification

All primes $p \leq 10^9$ were sieved (Eratosthenes, NumPy, run time ∼30 s). Prime count: 50 847 534. Twin-prime count (smaller members): 3 424 506.

6.1 (TPB) verified to $x = 10^9$

Writing $r_k = T_{k+1}/T_k$:

scope	$\sup r_k$	at T_k
all twins	2.200000	$5 \to 11$
$T_k > 100$	1.280374	107
$T_k > 10^3$	1.081880	1 319
$T_k > 10^6$	1.000711	1 122 281

The only k with $r_k \geq 2$ is $T_k = 5$, $T_{k+1} = 11$; for every $T_k \geq 11$ we have $r_k < 2$ strictly. By Proposition **??**, this verifies (TPB) for all x in $[11, 10^9]$. The top twenty ratios (Table **??**) all occur at $T_k < 1000$.

Table 1: Top 20 ratios $r_k = T_{k+1}/T_k$.

rank	T_k	T_{k+1}	r_k	gap
1	5	11	2.2000	6
2	17	29	1.7059	12
3	3	5	1.6667	2
4	11	17	1.5455	6
5	41	59	1.4390	18
6	71	101	1.4225	30
7	29	41	1.4138	12
8	107	137	1.2804	30
9	659	809	1.2276	150
10	347	419	1.2075	72
11	59	71	1.2034	12
12	149	179	1.2013	30
13	881	1019	1.1566	138
14	197	227	1.1523	30
15	461	521	1.1302	60
16	239	269	1.1255	30
17	311	347	1.1158	36
18	281	311	1.1068	30
19	521	569	1.0921	48
20	137	149	1.0876	12

6.2 Geometric confirmation: angle-records

All consecutive prime pairs with $p_n < 10^8$ (the first 5 761 455 primes) were examined; the running maximum of $\rho(p_n, p_{n+1})$ was tracked and each new maximum recorded.

- total angle-records: 440 312;
- twin-prime records (gap = 2): 440 311;
- non-twin records: 1, namely $(2, 3)$ with $\alpha = 33.6901°$.

By Theorem **??**, this is a geometric confirmation of (TPB) for the same range. The record angle increases monotonically from 35.5377° at the first twin $(5, 7)$ to 44.99999943° at the last record below 10^8.

6.3 Twin-gap growth law

Let $G_k = T_{k+1} - T_k$. Fitting $G_k = A(\log T_k)^\beta$ by least squares on $\log G_k$ vs. $\log\log T_k$:

fit range	β	A
$T_k > 10^3$	1.8633	0.7035
$T_k > 10^5$	1.8659	0.6981
$T_k > 10^7$	1.8664	0.6969

The exponent stabilizes at $\beta \approx 1.866$ across fit ranges, below the Hardy–Littlewood heuristic prediction $\beta = 2$. This suggests an empirical finite-range correction to the HL leading order in the average gap; whether $\beta \to 2$ as $T_k \to \infty$ is an open empirical question that would require data beyond 10^{10} to resolve.

6.4 Extreme-gap overshoot

Define the per-decade overshoot

$$\Omega(e) := \frac{\max\{G_k : T_k \in [10^e, 10^{e+1})\}}{(\log T_k^\star)^2},$$

where $T_k^\star$ is the argmax.

Table 2: Per-decade twin-gap statistics.

decade	n twins	mean G	max G	at T_k	$(\log T)^2$	Ω
$[10^1, 10^2)$	6	15.00	30	71	18.2	1.651
$[10^2, 10^3)$	27	34.00	150	659	42.1	3.560
$[10^3, 10^4)$	170	52.87	210	5 879	75.3	2.788
$[10^4, 10^5)$	1 019	88.46	630	62 297	121.9	5.169
$[10^5, 10^6)$	6 945	129.57	1 452	850 349	186.4	7.789
$[10^6, 10^7)$	50 811	177.13	1 722	9 923 987	259.6	6.635
$[10^7, 10^8)$	381 332	236.01	2 868	96 894 041	338.2	8.481
$[10^8, 10^9)$	2 984 193	301.59	4 770	698 542 487	414.7	11.502

Ω grows approximately monotonically, roughly seven-fold across seven decades. Extreme twin-gaps exceed $(\log T)^2$ by a slowly growing factor; this is the twin-prime analog of the Cramér–Granville phenomenon for ordinary primes. Note that this extreme behavior does *not* threaten (TPB), since $(\log T_k)^2/T_k \to 0$ exponentially faster than the relevant $T_k \cdot 1$ scale; the growing Ω affects only the tightness of any sharpened form of (TPB).

7 Twin-Ramanujan Primes and the Dyadic Twin-Prime Threshold

The (TPB) inequality $\pi_2(2x) - \pi_2(x) \geq 1$ asks for *at least one* twin prime in every dyadic window past a cutoff. A natural strengthening is to ask for *at least* n twin primes in every dyadic window. This yields a family of thresholds in direct analogy with the classical Ramanujan primes.

7.1 Classical Ramanujan primes

Ramanujan [**?**] proved that for every integer $n \geq 1$, the inequality $\pi(x) - \pi(x/2) \geq n$ holds for all sufficiently large x. The n-th Ramanujan prime is defined as

$$R_n \;=\; \min\Big\{\, R \,:\, \pi(x) - \pi(x/2) \geq n \text{ for all } x \geq R \,\Big\}, \tag{2}$$

so that $R_1 = 2$ (Bertrand's postulate itself). The first several values are $R_n = 2, 11, 17, 29, 41, 59, 67, 71, \ldots$ (OEIS `A104272` [**?**]). Sondow [**?**] established the asymptotic $R_n \sim p_{2n}$ (twice the n-th prime) and initiated the modern study of this sequence; later work extended the construction to c-Ramanujan primes and related objects [**?**, **?**].

7.2 The twin analog

We define, in direct parallel,

$$R_n^{\text{twin}} \;=\; \min\Big\{\, R \,:\, \pi_2(x) - \pi_2(x/2) \geq n \text{ for all } x \geq R \,\Big\}, \tag{3}$$

the least cutoff past which every dyadic interval $(x/2, x]$ contains at least n twin primes. By construction, R_n^{twin} is non-decreasing in n, and (TPB) is exactly the statement that R_1^{twin} exists and is equal to 11.

Theorem 9. $R_1^{\text{twin}} = 11$.

Proof (restatement of (TPB) *verification).* The function $f(x) := \pi_2(x) - \pi_2(x/2)$ is piecewise constant with a $+1$ jump at each twin prime $x = T_k$ and a -1 jump at each $x = 2T_k$. Direct evaluation on the first few twins shows $f(x) = 0$ on $[10, 11)$ (because $\pi_2(10) = 2$ and $\pi_2(5) = 2$), whereas $f(11) = \pi_2(11) - \pi_2(5.5) = 3 - 2 = 1$. A finite check for $x \in [11, 1000]$ confirms $f(x) \geq 1$ throughout, and the verification of (TPB) for $x \leq 10^{10}$ in Section **??** extends this to all tested x. Hence the infimum in (**??**) with $n = 1$ is 11. □

7.3 Computed values

From the twin-prime list up to 10^{10}, we computed R_n^{twin} via an event walk on the 42 million ± 1 transitions of f. Every R_n^{twin} lies within the data-certified range when $n \leq f(10^{10}) = 12{,}794{,}513$. Because f at the top of our range exceeds this maximum n by construction, each computed value is a true threshold, not merely an upper bound.

Table 3: First 30 twin-Ramanujan primes.

n	R_n^{twin}	n	R_n^{twin}	n	R_n^{twin}
1	11	11	821	21	1787
2	59	12	1019	22	1871
3	101	13	1049	23	1877
4	149	14	1061	24	1931
5	179	15	1289	25	1949
6	227	16	1319	26	2081
7	569	17	1427	27	2129
8	599	18	1451	28	2237
9	641	19	1481	29	2657
10	809	20	1667	30	2687

Remark 10 (Every R_n^{twin} is a twin prime)**.** *The function f changes by ± 1 at isolated events: $+1$ at each twin prime T_k and -1 at each $x = 2T_k$. A transition of f from strictly below n to at least n can only occur at a $+1$ event, i.e., at a twin prime. Since R_n^{twin} is defined as the smallest R past which $f \geq n$ is sustained, it coincides with the first twin prime at which such a sustained transition is achieved. In particular, every R_n^{twin} is itself a twin prime. We verified this directly for all $n \leq 10^6$ in our data: all one million computed values are twin primes.*

Table 4: Larger R_n^{twin} at logarithmic scales.

n	R_n^{twin}
100	14,009
1 000	217,337
10 000	3,241,487
100 000	45,529,751
1 000 000	computed in `data/twin_ramanujan_primes.csv`

7.4 Growth rate

Sondow's asymptotic for the classical sequence is

$$R_n \sim p_{2n} \sim 2n \log n \qquad (n \to \infty). \tag{4}$$

The heuristic analog for twins comes directly from the Hardy–Littlewood prediction $\pi_2(x) \sim 2C_2\, x/(\log x)^2$: writing $R = R_n^{\text{twin}}$ and requiring $\pi_2(R) - \pi_2(R/2) \approx n$, we obtain $n \approx 2C_2\, R/(2(\log R)^2)$, hence

$$R_n^{\text{twin}} \sim \frac{n\,(\log n)^2}{C_2} \qquad (n \to \infty), \tag{5}$$

a factor of roughly $\log n/(2C_2) \approx 0.76 \log n$ times the ordinary R_n. On the data range considered here, the observed ratio R_n^{twin}/R_n grows slowly with n, consistent with (**??**) up to the finite-size corrections documented in Section **??**.

7.5 Positioning

Theorem **??** recasts (TPB) as the statement $R_1^{\text{twin}} = 11$, placing it inside the Ramanujan-prime family of dyadic-density thresholds. The classical Ramanujan primes quantify how large x must be before every dyadic interval contains n *ordinary* primes; the twin-Ramanujan primes ask the same question for twin primes. The construction is the immediate analog of Ramanujan's original definition, but replaces π by π_2; c-Ramanujan primes [**?**] generalize in a different direction (varying the interval ratio c for the ordinary-prime count). To our knowledge, the specific sequence R_n^{twin} has not previously been defined or tabulated; we propose it as the natural dyadic-density sequence attached to twin primes, and the corresponding data file (first 10^6 values) as a candidate for inclusion in the OEIS.

The same construction applies to any admissible pair-constellation $\mathcal{C} = \{0, g\}$, yielding $R_n^{\mathcal{C}}$. The first values for $\mathcal{C} \in \{\{0,4\},\{0,6\}\}$ are $R_1^{\{0,4\}} = 7$ and $R_1^{\{0,6\}} = 5$, giving the cousin and sexy thresholds a geometric interpretation; this is developed in PG III as the Generalized Bertrand Principle.

8 Related Work

The study of twin primes is extensive. We summarize the most directly relevant strands.

Ramanujan primes. Ramanujan's 1919 theorem [**?**] that $\pi(x) - \pi(x/2) \geq n$ holds for all $x \geq R_n$ is the direct parent of (TPB). Sondow [**?**] modernized the theory with the asymptotic $R_n \sim p_{2n}$; Amersi–Beckwith–Miller–Ronan–Sondow [**?**] generalized to c-Ramanujan primes (varying the

interval ratio), and Paksoy [**?**] introduced derived Ramanujan primes. Sondow–Nicholson–Noe [**?**] studied which Ramanujan primes happen to be twin primes (a different object from our R_n^{twin}: they examine ordinary Ramanujan primes that fall within twin-prime pairs, whereas R_n^{twin} is the dyadic-density threshold for π_2 itself).

The Hardy–Littlewood conjecture. Hardy and Littlewood conjectured in 1923 that the number of twin primes up to x is asymptotically $2C_2 \int_2^x dt/(\log t)^2$, with C_2 the twin-prime constant. No non-trivial lower bound of this form, or even of the form $\pi_2(x) \gg x/(\log x)^A$, is known unconditionally.

Bounded gaps. Zhang's 2013 theorem established that $\liminf_{n\to\infty}(p_{n+1}-p_n) < \infty$ with a bound of 7×10^7. The Polymath8 collaboration reduced this to 4 680 using refinements of Zhang's method, and to 246 using Maynard's multi-variable sieve (Polymath8b). Under the Elliott–Halberstam conjecture, the bound drops to 12, and under the generalized Elliott–Halberstam conjecture it reaches 6. None of these results give a twin prime (gap 2) in *every* interval $(x, 2x]$; they give a bounded-gap pair at *some* scales infinitely often.

Explicit bounds on π_2. Upper bounds of Brun-sieve and Selberg-sieve type give $\pi_2(x) \ll x/(\log x)^2$ unconditionally, which is compatible with HL up to a constant factor. Explicit constants are known in the sieve literature; they are not sharp enough to yield (TPB) unconditionally in any finite range beyond direct computation.

Twin primes in short intervals. Classical results (Heath-Brown, Friedlander-Iwaniec, and others) give conditional short-interval estimates of the form $\pi_2(x+y) - \pi_2(x) > 0$ for $y \geq x^{1-\delta}$ under various hypotheses; however these do not reduce y to x (the dyadic scale) unconditionally. We are not aware of a statement in the literature that isolates the dyadic inequality $\pi_2(2x) \geq \pi_2(x)+1$ as a target in its own right.

Summary. (TPB) occupies a gap in the literature: it is natural, elementary to state, verifiable by direct computation, and believed true, yet it has not been articulated explicitly as a conjecture. Its provability under Hardy–Littlewood is immediate; its unconditional proof appears to require either a stronger short-interval result than is currently available or a new sieve-theoretic input.

9 Conclusion

(TPB) is a clean, elementary, unproven conjecture about twin primes that sits strictly between the Zhang–Maynard bounded-gaps theorems and the full twin-prime conjecture. It is equivalent to the ratio statement $T_{k+1} < 2T_k$ for $T_k \geq 11$, equivalent to the geometric statement that every angle-record of the Prime Triangle sequence with $p_n \geq 3$ is a twin prime, and equivalent to the identity $R_1^{\text{twin}} = 11$ within the Ramanujan-prime family.

The four formulations are mutually reducible via the elementary $2P$-beats lemma and the Ramanujan-threshold construction; each provides a different avenue for verification and potential proof. Our computations verify (TPB) for all $x \leq 10^9$, a range that dwarfs the scales at which the ratio envelope tightens to $r_k < 1.001$, and produce the twin-Ramanujan sequence R_n^{twin} for n up to 12 794 513.

We have also observed that the average twin-gap is well-fitted by $0.70(\log T_k)^{1.866}$ and that the extreme-gap overshoot $\max G_k/(\log T_k)^2$ grows monotonically across decades. Whether the

exponent β converges to 2 asymptotically, and whether the overshoot growth follows a $(\log\log T)$ or power-of-$\log\log T$ law, are open empirical questions.

A near-term goal is the unconditional proof of (TPB) by a sieve-theoretic argument adapting the Maynard multi-dimensional sieve to the dyadic short-interval regime; a complementary goal is the extension of the empirical verification to 10^{12} and beyond. We leave both to future work.

Data availability

All computations reproduce from `scripts/pg_twin_angle_analysis.py` and `scripts/pg_twin_ramanujan.py`. The derived datasets (`twins_1e9.npy`, `twins_1e10.npy`, `angle_records_1e8.csv`, `ratio_top20.csv`, `decade_gap_table.csv`, `twin_ramanujan_primes.csv`) are stored alongside.

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